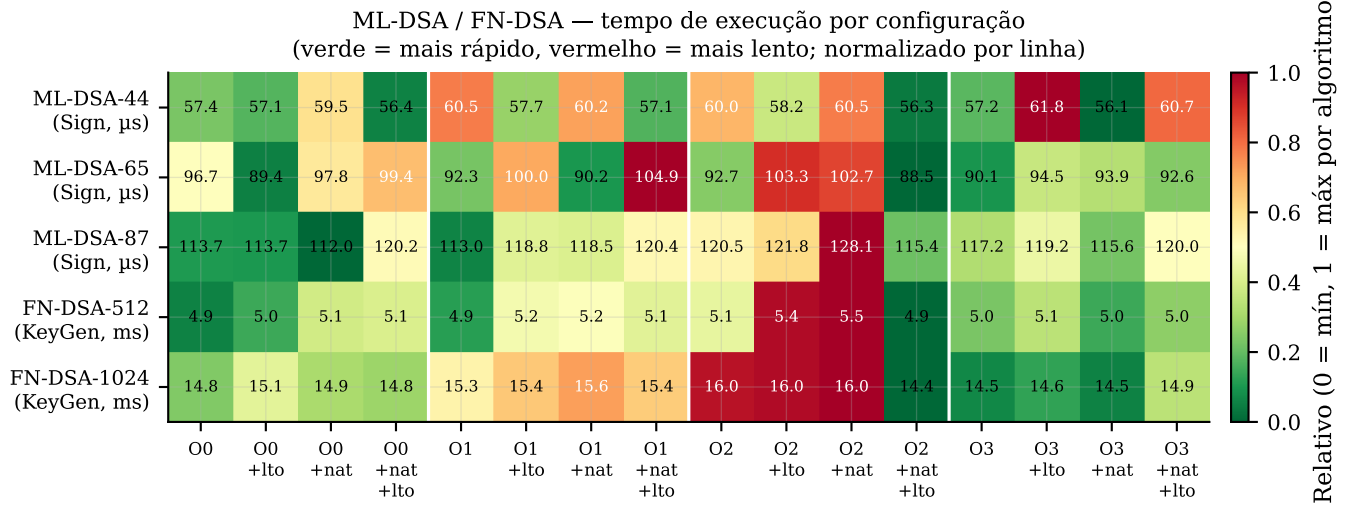
ML-DSA / FN-DSA — tempo de execução por configuração
(verde = mais rápido, vermelho = mais lento; normalizado por linha)
ML-DSA-44 (Sign, µs)
57.4 57.1 59.5 56.4 60.5 57.7 60.2 57.1 60.0 58.2 60.5 56.3 57.2 61.8 56.1 60.7
ML-DSA-65 (Sign, µs)
96.7 89.4 97.8 99.4 92.3 100.0 90.2 104.9 92.7 103.3 102.7 88.5 90.1 94.5 93.9 92.6
ML-DSA-87 (Sign, µs)
113.7 113.7 112.0 120.2 113.0 118.8 118.5 120.4 120.5 121.8 128.1 115.4 117.2 119.2 115.6 120.0
FN-DSA-512 (KeyGen, ms)
4.9 5.0 5.1 5.1 4.9 5.2 5.2 5.1 5.1 5.4 5.5 4.9 5.0 5.1 5.0 5.0
FN-DSA-1024 (KeyGen, ms)
14.8 15.1 14.9 14.8 15.3 15.4 15.6 15.4 16.0 16.0 16.0 14.4 14.5 14.6 14.5 14.9
O0
O0 +lto
O0 +nat
O0 +nat +lto
O1
O1 +lto
O1 +nat
O1 +nat +lto
O2
O2 +lto
O2 +nat
O2 +nat +lto
O3
O3 +lto
O3 +nat
O3 +nat +lto
Relativo (0 = mín, 1 = máx por algoritmo)
1.0
0.8
0.6
0.4
0.2
0.0